



ORACLE

Cloud Operations for ExaDB-D

Maintenance Operations Scenarios

Operations Advisory Team

Office of CTO | EMEA Solution Engineering

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Agenda

- Introduction
- Maintenance Operations Scenarios
 - Schedule infrastructure maintenance
 - Update/Upgrade VM Guest
 - Update/Upgrade Grid Infrastructure Update/Upgrade
 - Update Database
 - Updating Cloud Tooling
 - Gather Troubleshooting Information
 - Manage storage cells information

Introduction

Introduction



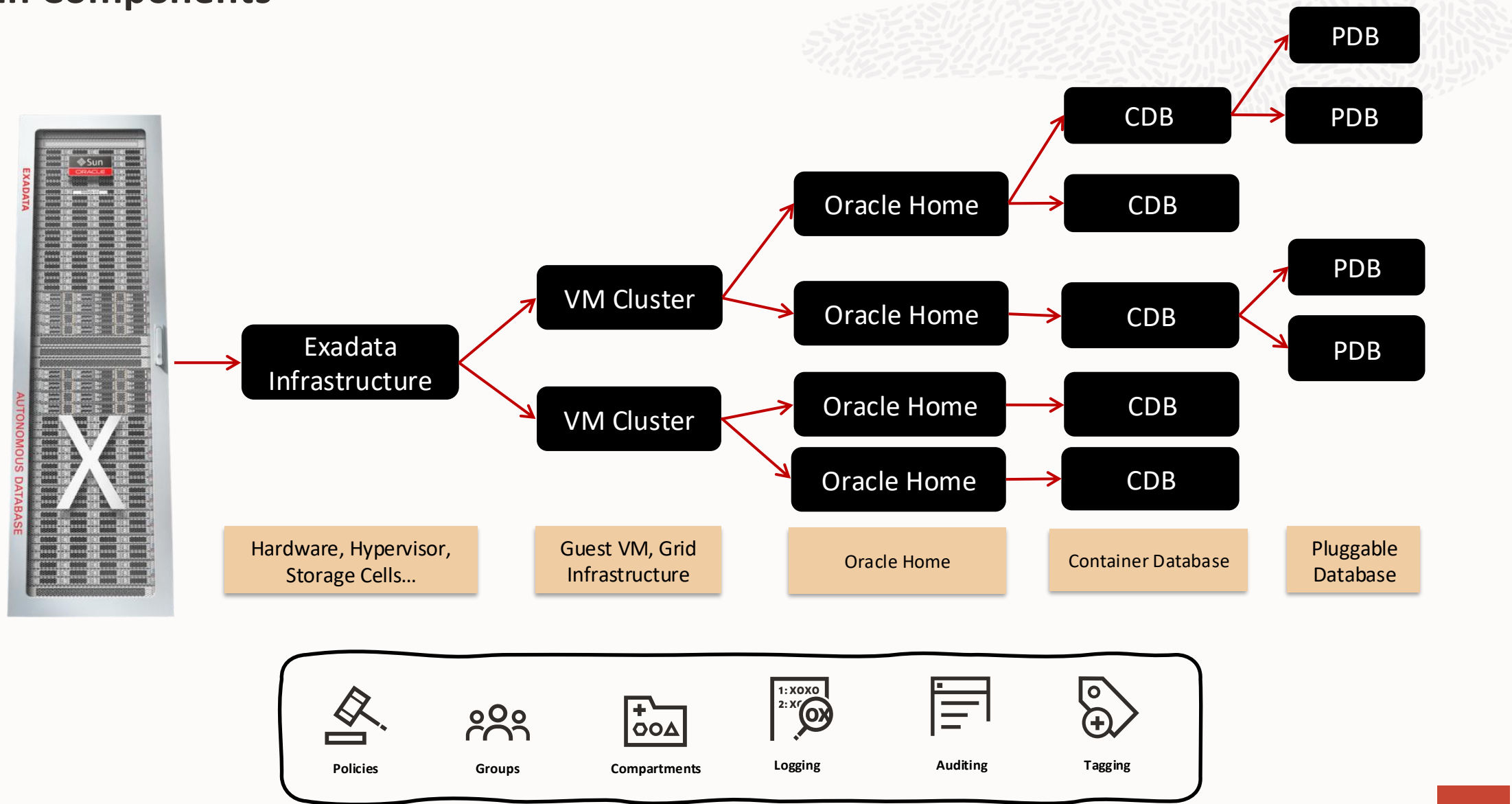
This Workshop provides information about Exadata Database Service on Dedicated Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VMC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

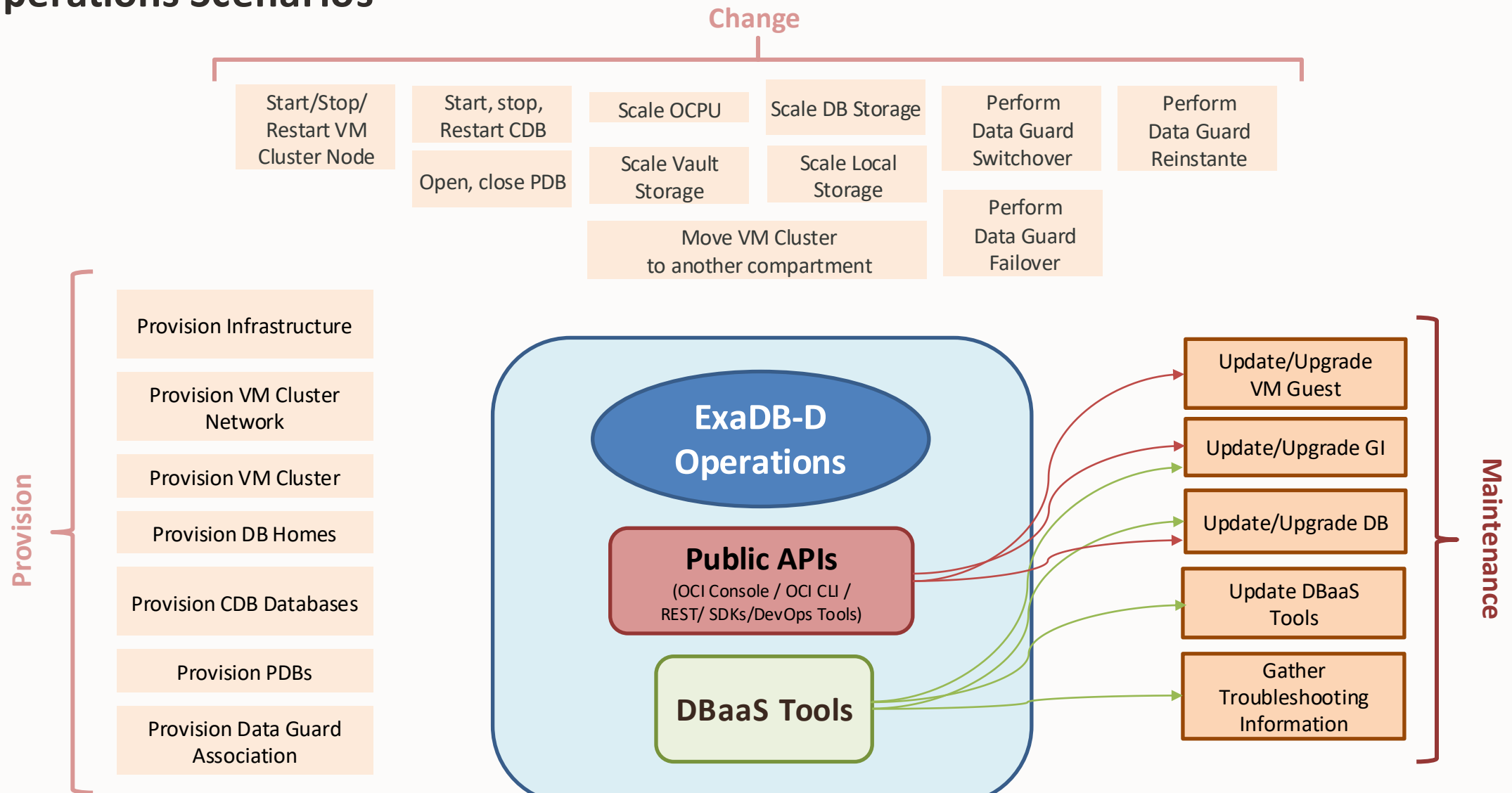
To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

Main Components



Maintenance Operations

Operations Scenarios



Schedule infrastructure maintenance

Infrastructure Update Types



Quarterly Infrastructure Maintenance

Comprehensive updates that may include one or more of the following:



New Exadata Infrastructure image



Latest security or other essential patches for infrastructure components



Updates to infrastructure components (DB Servers, storage servers, network switches)



Rolling maintenance to preserve availability



Monthly Security Maintenance

Critical security updates applied more frequently to protect your environment.



Security and product fixes addressing vulnerabilities across all CVSS severity levels



Online patching of DB Servers using Ksplice (no reboot)

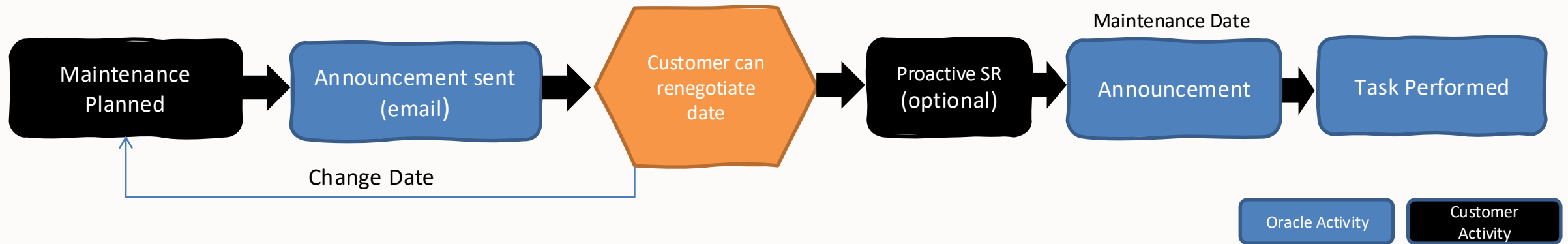


Rolling updates for Storage Servers



Designed to deliver critical fixes while maintaining high availability

Planned Quarterly Infrastructure Maintenance Workflow



- Oracle is responsible to perform the maintenance
- Planned maintenances may need to perform VM Node reboots
- Customer is responsible of the components running in the VM
- In case of needing to report issues while the maintenance open an SR
- There is a maintenance per quarter: **Customer might skip some maintenances per year**

Infrastructure Maintenance Quarterly Process



Database Servers

- **Database server maintenance is done in a rolling manner by default.**
- Updates are applied to the Exadata database server hosts.
- Customer-managed VMs are restarted during maintenance but are not directly patched as part of infrastructure maintenance.
- Database connections are drained prior to shutting down each database server.
- Database server maintenance is completed before storage server maintenance begins.



Storage Servers

- **Storage server maintenance is also done in a rolling manner by default.**
- Post-database servers updates, storage servers are updated sequentially to maintain availability.
- Storage server updates do not impact database server availability.
- Data remains protected through Oracle-managed storage redundancy while storage components are updated in a rolling manner.

Quarterly Maintenance Controls and Scheduling

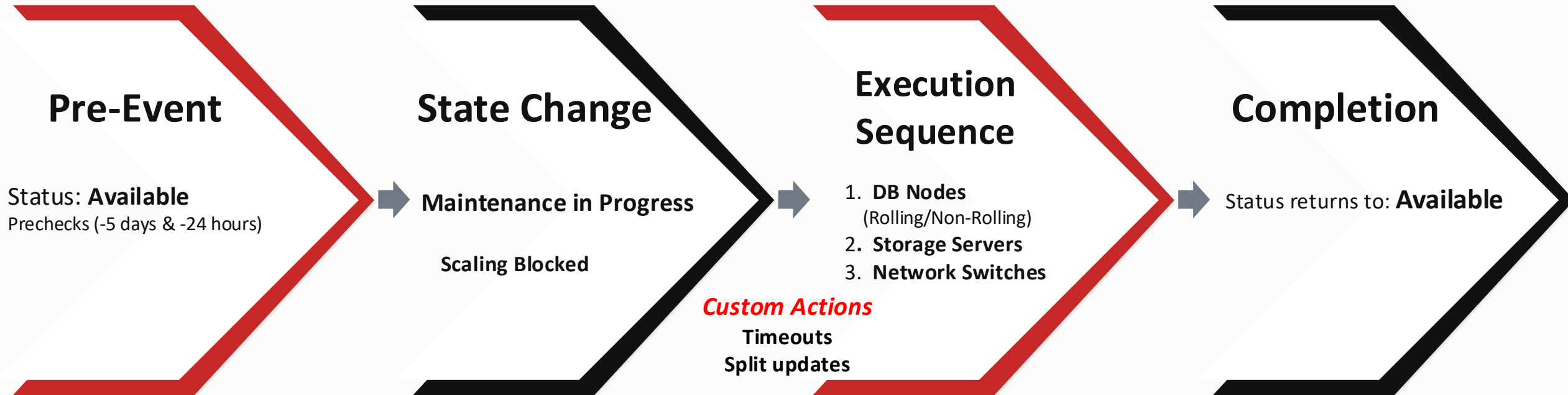
CUSTOM ACTIONS

- Enables customer tasks before each node is patched
- Configurable timeout (15–120 min) and real-time options to Resume or Extend via Console/API.
 - Rolling Mode: Pauses before each DB server update.
 - Non-Rolling Mode: Pauses once before updating all DB servers.

SCHEDULING FLEXIBILITY & MAINTENANCE PLANS

- Define maintenance scheduling policies aligned with business cycles (months/weeks/days).
- Split updates into separate windows (DB servers on Saturday, storage on Sunday, or distribute across days).
- Enforce Window Duration: If maintenance overruns, Oracle pauses and reschedules tasks to prevent business-hour impact.
- Additional maintenance windows can be added to a Scheduling Policy.
- Maintenance windows in a scheduling policy should be no more than 3 weeks apart.

The Maintenance QMR Event Lifecycle



Scaling operations (CPU/Storage) are blocked during the entire maintenance sequence.

Monthly Security Maintenance Process



Include fixes for critical security vulnerabilities across all CVSS severity levels and critical product issues.



Scheduled monthly during a 21-day window starting between the 18th and 21st and runs until the 9th – 12th of the following month.



Oracle sends a schedule **notification at least 7 days prior** to the maintenance start
You **typically receive at least 2 weeks' notice** to plan for these updates.

If a **DB restart is required**, you get **≥ 2 weeks notice**



Can be rescheduled within the 21-day window but cannot be skipped.



Database Servers



Applied **automatically online** using *Ksplice* technology.



No reboots required for physical hosts or Guest VMs.



Databases remain available during maintenance.



Storage Servers

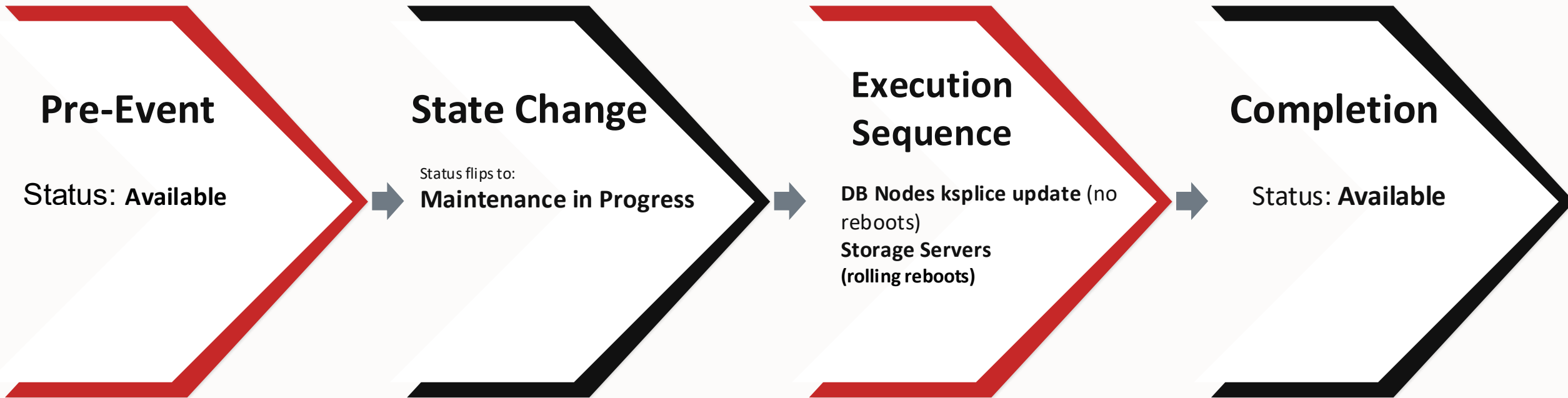


Applied in a **rolling manner** (one server at a time).



Designed to have minimal to no impact on database availability.

The Maintenance SMR Event Lifecycle



CPU scaling and VM startup/shutdown are the only operations supported during monthly infrastructure maintenance.

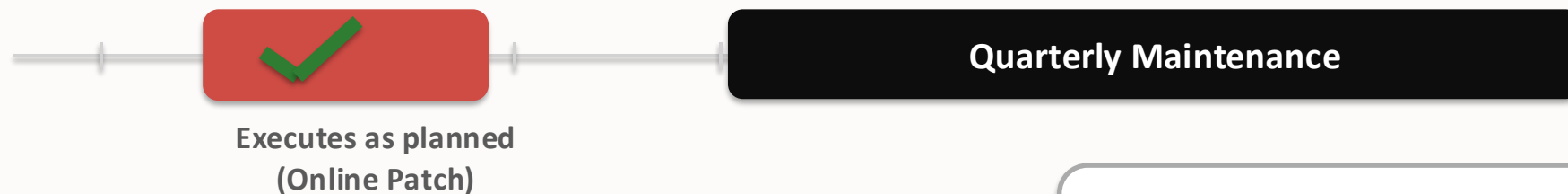
Managing the Intersection of Monthly and Quarterly Cycles

The 24-Hour Rule: Quarterly updates are cumulative and include previous monthly security fixes.

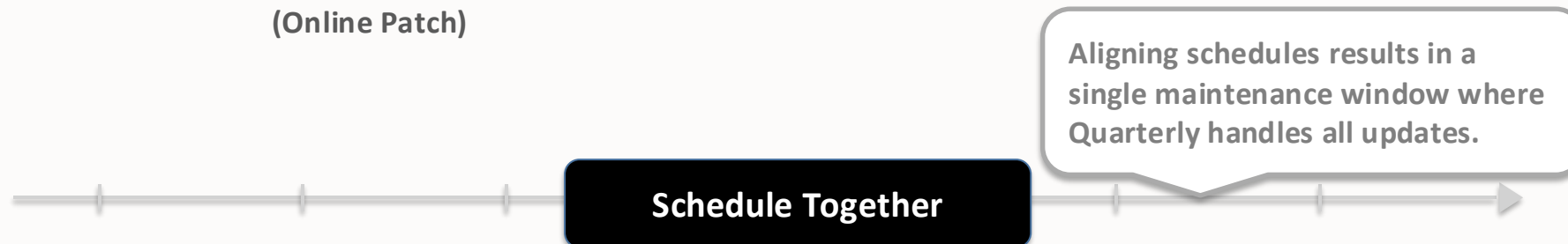
Scenario A: Overlap 0-24 Hours



Scenario B: Separated > 24 Hours



Recommendation: Best Practice



Update/Upgrade VM Guest

Updating an Exadata Cloud VM Cluster Best Practices

Exadata VM Cluster Image Updates let you automatically update the OS image on VM cluster nodes directly from the OCI Console or APIs.

Execution Strategy & Tools

- **Prioritize automation:** Use OCI Console, or OCI CLI instead of manual OS tools to keep control-plane metadata in sync.
- **Patching order:** Patch the VM Cluster (GI + OS) before Database Homes and databases.
- **Avoid manual changes:** Oracle-required users and automation keys should not be altered or deleted; use supported workflows (for example, UpdateVmCluster) for SSH key management.

Preparation & Pre-Patching Validation

- **Protect & validate first:** Back up databases, then run Precheck immediately before patching (≥15 GB free in /u01, Clusterware up, all nodes up). Optionally run a standalone Precheck earlier to surface issues in advance.
- **Version compatibility:** Verify target versions in “Exadata Cloud Service Software Versions” (Doc ID 2333222.1) and confirm OEM/plugin compatibility (Doc ID 1626579.1).
- **Object Storage readiness:** Ensure OCI Object Storage access (incl. Swift endpoint connectivity); Oracle recommends using a Service Gateway in the VCN and reviewing the required security rules for ExaDB-D.
- **Custom software caveat:** Third-party/custom RPMs may be removed to satisfy dependencies; reinstall target-OS-compatible versions after. Kernel-tainting software is generally not allowed.
- **Operational impact:** During patching, resources can show Updating and some lifecycle operations may be temporarily unavailable; schedule changes outside the window.



OCI Control Plane Interfaces

OCI Console

*Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → VM Clusters →
Select VM Cluster → Updates (OS) → Select Actions (3 dots) on the VM OS Update → Precheck/Apply Exadata OS Image Update*

Command Line Interface (OCI CLI)

Lists the maintenance updates

```
oci db cloud-vm-cluster list-updates --cloud-vm-cluster-id <vm-cluster-ocid> --update-type OS_UPDATE[OPTIONS]
```

Gets information about a specified maintenance update

```
oci db cloud-vm-cluster get-update --cloud-vm-cluster-id <vm-cluster-ocid> --update-id <update-ocid>
```

Updates the specified Exadata VM cluster on ExaDB-D Infrastructure

```
oci db cloud-vm-cluster update --cloud-vm-cluster-id <vm-cluster-ocid> \  
--update-id <update-ocid> --update-action <PRECHECK|ROLLING_APPLY|NON_ROLLING_APPLY|ROLLBACK>
```

If an Exadata infrastructure maintenance operation is scheduled to start within 24 hours, the Exadata Image update feature becomes unavailable.

Software Images

Software Images

Software images are resources within your tenancy that you create before provisioning or updating a Exadata Cloud Infrastructure instance, Database Home, database, or Grid Infrastructure.

Types

- Grid Infrastructure
- Database

Procedure

- Create custom Database/GI images based on Software Images, RU, and one-off (interim) patches.
- Provision a new Database/GI home with the desired RU and one-off (interim) patches.
- Out of place Database/GI update

Tooling Options

- Oracle Cloud Web based: OCI Console, OCI-CLI, REST API, SDK
- Guest VM Command Line Interface (CLI): dbaascli
- IaC: Terraform
- OCI Managed Services: Exadata Fleet Update

Software Images



RECOMMENDED

- Ensures consistent, standardized patch levels across environments.
- Reduces patching time by deploying pre-patched images.
- Avoids manual intervention during patch apply.
- Minimizes human error during patching and configuration.
- Simplifies rollback by allowing quick switch-back to the previous Oracle Home or fast redeployment of previous images.
- Eases large-scale updates as can be used combined with other services as Exadata Fleet Update

Update/Upgrade Grid Infrastructure

Update/Upgrade Grid Infrastructure

Requirements

- Use the OCI Management Interfaces to perform update/patching operations.
- Ensure all servers and database instances are up and running.
- Ensure /u01 (GI Home directory) has 15GB of free space before a Grid Infrastructure (GI) update.
- Ensure /u02 (DB Home directory) has 15GB free space before a DB Home update.
- Ensure to update the Grid Infrastructure before updating or creating a DB Home:
- The Grid Infrastructure must be at the same or higher version than the database version you want to patch to.

Recommendations

- Back up your databases before you apply any updates to the DB Home.
- Before you apply any update, run the precheck operation.
- Run the patching precheck function in advance of maintenance window to validate system readiness.
- Plan to run the patch operations at a time when they will have minimal impact on users.
- Use out-of-place patching for database patching.

How can you update/upgrade the GI?

3 +1 ways :

- From the Console
- With [OCI CLI](#)
- With [dbaascli: how-tos](#)
- OCI Managed Services: Exadata Fleet Update

In all cases you can run a precheck or apply operation in rolling.

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → VM Clusters →
Select VM Cluster → Updates (GI) → Select Actions (3 dots) on the DB System Patch → **Precheck/Apply Grid Infrastructure Update**

Command Line Interface (OCI CLI)

Lists the GI versions available

```
oci db cloud-vm-cluster list-updates --cloud-vm-cluster-id <cloud_vm_cluster_ocid> --update-type GI_PATCH|GI_UPGRADE
```

Inspect the selected patch

```
oci db cloud-vm-cluster get-update --cloud-vm-cluster-id <cloud_vm_cluster_ocid> --update-id <update_ocid>
```

GI Precheck

```
oci db cloud-vm-cluster update --cloud-vm-cluster-id <cloud_vm_cluster_ocid> --update-id <update_ocid> --update-action PRECHECK
```

Apply the GI patch/upgrade

```
oci db cloud-vm-cluster update --cloud-vm-cluster-id <cloud_vm_cluster_ocid> --update-id <update_ocid> \  
--update-action ROLLING_APPLY
```

Local VM Command-Line Interfaces - dbaascli

dbaascli would give you more control over the process, choosing just to patch an instance and having the possibility to check logs or perform health check between nodes.

Upgrade/Update GI

List available updates:

```
# dbaascli cswlib showImages --product grid
```

GI Patching precheck:

```
# dbaascli grid patch --targetversion <version to patch> --executePrereqs
```

GI Patching:

```
# dbaascli grid patch --targetversion <version to patch> --nodeList <hostname1> --waitForCompletion
```

```
# dbaascli grid patch --targetversion <version to patch> --nodeList <hostname2> --waitForCompletion
```

GI Patching using a Software Image:

```
# dbaascli grid patch --targetVersion <target_software_version_number> --createImage
```

```
# dbaascli grid patch --targetVersion <target_software_version_number> --imageLocation <location_of_patched_software_image>
```

GI Patching:

```
# dbaascli gridHome create --version <target_software_version_number>
```

```
# dbaascli grid patch --targetHome <location_of_patched_GI_home >
```

Logs of the tool can be seen in:

```
/var/opt/oracle/log/
```

Update/Upgrade Database

Database patching options

***Database in-place patching** is the process of applying patches directly to an existing Oracle Database home, updating the software in that same location without creating or switching to a new Oracle home.*

***Database out-of-place patching** is the process of applying patches to a new Oracle Home and then switching the database to that updated home, leaving the original home unchanged.*

Best practices

- Back up your databases before you apply any patches.
- Patch a VM cluster before you patch the Databases Homes and databases on that resource.
- Run the precheck operation to ensure your VM cluster or Database Home meets the requirements for that patch.
- Use Software Images.
- Use **out-of-place** patching if possible:
 - Less downtime
 - Easier roll back the database
 - Parallelism is allowed using **dbaascli --allowParallelDBMove**

FAQ

- Is better to use dbaascli, OCI CLI or the OCI Console?
- Is better in-place or out-of-place patching approach?

OCI Control Plane Interfaces

OCI Console

*Menu → Oracle AI Database → Oracle Exadata Database Service on Dedicated Infrastructure → VM Clusters →
Select VM Cluster → Container databases → Select the Container database → More actions → **Move to another home / Upgrade***

Command Line Interface (OCI CLI)

List Patches

```
# oci db patch list-db-home --db-home-id <home_id>
```

In-place: Patch existing DB home with new binaries version

```
# oci db db-home update --db-home-id <DB_HOME_OCID> --database-software-image-id <IMAGE_OCID>
```

Out-of-place: Move the DB to a new home previously created

```
# oci db database update --database-id <DATABASE_OCID> --db-home-id <NUEVO_DB_HOME_OCID> --wait-for-state AVAILABLE
```

Check Work Request

```
# oci work-requests work-request list --compartment-id <COMPARTMENT_OCID> --resource-id <DATABASE_OCID> --all --output table
```

Local VM Command-Line Interfaces - dbaascli

Move database to another home (out-of-place)

Precheck

```
# dbaascli database move --dbname <DBNAME> --oracleHome <TARGET_OH_PATH> --executePrereqs
```

Run in Background (with datapatch)

```
# dbaascli database move --dbname <DBNAME> --oracleHome <TARGET_OH_PATH> --waitForCompletion false
```

Run in Background (without datapatch)

```
# dbaascli database move --dbname <DBNAME> --oracleHomeName <TARGET_HOME_NAME> --skipDatapatch
```

Run precheck & dataPatch

```
# dbaascli database runDatapatch --dbname <DBNAME> --executePrereqs
```

```
# dbaascli database runDatapatch --dbname <DBNAME>
```

In-place

```
# dbaascli dbHome patch --oracleHome <CURRENT_HOME_PATH> --targetVersion <TARGET_VERSION> --executePrereqs
```

```
# dbaascli dbHome patch --oracleHome <CURRENT_HOME_PATH> --targetVersion <TARGET_VERSION> --nodes <hostname1> --waitForCompletion
```

```
# dbaascli dbHome patch --oracleHome <CURRENT_HOME_PATH> --targetVersion <TARGET_VERSION> --nodes <hostname2> --waitForCompletion
```

Apply the patch (WITHOUT datapatch)

```
# dbaascli dbHome patch --oracleHome <CURRENT_HOME_PATH> --targetVersion <TARGET_VERSION> --skipDatapatch
```

```
# dbaascli dbHome patch --oracleHome <CURRENT_HOME_PATH> --targetVersion <TARGET_VERSION> --skipDatapatchForDB <db1,db2,...>
```

Logs of the tool can be seen in:

```
/var/opt/oracle/log/
```

Patching Data Guard Enabled Environments

Patching process

- Patch the standby DB software first, then patch the primary.
- Applying the patch on the primary also applies datapatch on the standby via redo transport.
- Oracle recommendation: out-of-place
 - If standby needs one-offs: build a Database Custom Software Image (includes one-offs) and deploy a new Oracle Home as part of the out-of-place flow.
 - If using in-place: apply one-offs manually to patch the Oracle Home.

Operational notes

- Out-of-place moves are serial by default (one DB at a time), but you can enable parallel moves with dbaascli database move.
- The patching process is rolling, regardless of in-place or out-of-place.
- The tooling detects Data Guard and does not run datapatch on the standby.
- If you need to skip datapatch on the primary, use --skipDatapatch.
- Redo Apply is managed by Data Guard Broker; when patching via tooling, MRP handover across nodes is automatic, so you do not need to manually stop/move apply between nodes.
- Avoid stopping redo transport on the primary instances for updates.
- Verify that Data Guard transport and apply are healthy before/after the maintenance cycle.

Rollback/Resume

Rollback / Resume

- Resume and Rollback operations are supported for Oracle Home patching, Oracle Grid Infrastructure patching, and database move operations.
- While using the --resume or --revert command options, ensure that you use the same command from the same node that was used for actual create operation flow.
- You can resume database creation only if there is a failure in the post database creation step.

Resume / Rollback

Rollback de DB Home

```
# dbaascli dbHome patch --oracleHomeName <DB_HOME_NAME> --targetVersion <TARGET_VERSION> --resume  
# dbaascli dbHome patch --oracleHomeName <DB_HOME_NAME> --targetVersion <TARGET_VERSION> --rollback
```

Rollback de Grid Infrastructure

```
# dbaascli grid patch --targetVersion <TARGET_VERSION> --resume  
# dbaascli grid patch --targetVersion <TARGET_VERSION> --rollback
```

Rollback “database move”

```
# dbaascli database move --dbname <DBNAME> --oracleHomeName <TARGET_DB_HOME_NAME> --resume  
# dbaascli database move --dbname <DBNAME> --oracleHomeName <TARGET_DB_HOME_NAME> --rollback
```

Updating Cloud Tooling

Updating Cloud Tooling

In ExaDB-D, “cloud tooling” is the Oracle-provided software stack installed on the Guest VMs (DomU) that enables and orchestrates cloud-managed lifecycle operations from inside the VM.

How can you update the cloud tooling manually?

1. Connect to a virtual machine as the opc user.

2. Start a root user command shell:

```
$ sudo -s
```

3. To show the version of latest available cloud tooling stack to update, run the following command:

```
> dbaascli admin showLatestStackVersion
```

4. To update to the latest available cloud tooling release, run the following command:

```
> dbaascli admin updateStack
```

The command takes care of updating the cloud tooling release on all the nodes of the cluster.

The cloud tooling is automatically updated by Oracle when new releases are made available. You can update on-demand the cloud-specific tooling by downloading and applying a software package containing the updated tools.

For more details and other available options, refer to **dbaascli admin updateStack --help**

Gather Troubleshooting Information

Local VM Command-Line Interfaces - dbaascli

Collect diagnostic log

```
# dbaascli diag collect --startTime 2026-12-05T10:00:00 --endTime 2026-12-05T11:00:00 --destLocation /tmp/test/

DBAAS CLI version 24.1.1.0.0
Executing command diag collect --startTime 2026-12-05T10:00:00 --endTime 2023-12-05T11:00:00
Job id: 70b03e50-98cc-4c2b-9684-1f82070bac88
Session log: /var/opt/oracle/log/diag/collect/dbaastools_2026-12-14_09-45-17-PM_42856.log
Loading PILOT...
Session ID of the current execution is: 15
Log file location: /var/opt/oracle/log/diag/collect/pilot_2026-12-14_09-45-21-PM_43526
-----
..
----- DIAG COLLECT PLUGIN RESULT -----
{
  "collectedArchive with SHA256 CheckSum" : "{/tmp/test/diag_collect/artifacts_diag_cloudlogs_20231214-2143/diag_cloudlogs_20231214-2143_node1.zip=8a26cffcdd72c261660d4f736c615981856e357749d90751b94f3eda19a9a70}"
}
dbaascli execution completed
```

Healthcheck

```
# dbaascli diag healthcheck --destLocation /tmp/test

DBAAS CLI version MAIN
Executing command diag healthcheck --destLocation /tmp/test
INFO: Starting diag healthcheck
INFO: Collected diag logs at: /tmp/test/diag_cloudlogs_20260322-2250.tar.gz
```

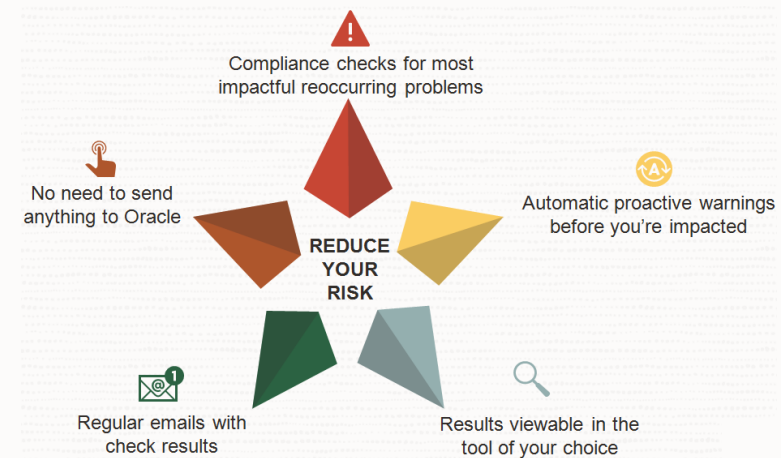
Automatic Health Framework (AHF)

\$ any user
as root

AHF helps you detect problems earlier and reduce the time to find root cause.

AHF is a collection of different tools as:

- **Health checks (EXAchk)** to compare your system with **best practices and commonly known issues**.
- **Log and diagnostic collection** with TFA (Trace File Analyzer), so you can gather the right files from all nodes and send one support bundle to Oracle Support.
- **Performance / hang diagnostics tools** (for example procwatcher) to help investigate slowdowns or hangs.
- **ExaWatcher** is an Exadata system **data collection + reporting utility**. It runs on each ExaDB-D server and captures very detailed OS + network statistics, plus cell-related statistics (for storage servers).



Examples

Verify tool status

```
# tfactl toolstatus
```

Run Exachk

```
# exachk
```

Run TFA diagnostic collection

```
$ tfactl diagcollect -from "MMM/dd/yyyy hh:mm:ss" -to "MMM/dd/yyyy hh:mm:ss"
```

Database Service Events

\$ any user
as root

Oracle Trace File Analyzer

```
# tfactl status
# tfactl start
# tfactl stop
```

sysLens

```
# systemctl status syslens
# systemctl is-enabled syslens
# /var/opt/oracle/syslens/bin/applin_ctl config_file get enable --file /etc/oracle/syslens/config/syslens.config
# tfactl get customerDiagnosticsNotifications
```

.-----.	
Hostname	
+-----+	
Configuration Parameter	Value
+-----+	
Send CEF notifications (customerDiagnosticsNotifications)	ON
'-----'	

Database Service Agent

```
# systemctl status dbcsagent.service
# systemctl start dbcsagent.service
```

Logs -> /opt/oracle/dcs/log/dcs-agent.log

Manage storage cells information

Managing Storage Cells information | ExaCLI

ExaCLI is a command-line tool used to gather information from Exadata Storage Servers (cells).

- In ExaDB-D there is a preconfigured user for Exadata Storage Server access called something like “cloud_user_\${clustername}”, where \$clustername is the name of the VM cluster being used

Gathering your user name to connect to Storage cells:

```
grid@node1 $ crsctl get cluster name
```

Getting password for “cloud_user_clustername” storage cell user:

```
opc@node1 $ $ /opt/exacloud/get_cs_data.py
```

Getting used storage cell servers:

```
root@node1# cat /etc/oracle/cell/network-config/cellip.ora
cell="192.168.136.5;cell="192.168.136.6"
cell="192.168.136.7;cell="192.168.136.8"
cell="192.168.136.9;cell="192.168.136.10"
```

Retrieve IORM plans:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e list iormplan [--cookie-jar]
```

Create a diagpack:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e create diagpack packStartTime=time, durationInHrs=3
```

Managing Storage Cells information | ExaCLI

ExaCLI command provides a subset of the on-premises Exadata command line utility

Show flashcache details:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e LIST flashcache detail
```

Show storage cells metrics for a particular object type:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e LIST METRICCURRENT attributes name, metricObjectName  
ORDER BY metricObjectName asc, name desc LIMIT 5
```

List SQL statements prevented from using Smart Scans:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e LIST QUARANTINE DETAIL
```

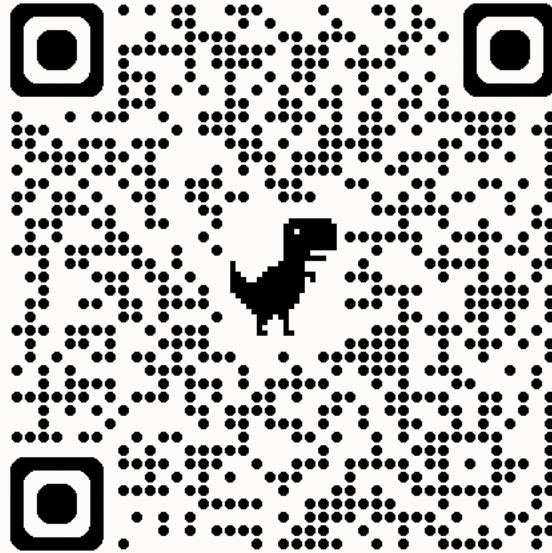
Show alerts:

```
opc@node1 $ exacli -c cloud_user_clustername@192.168.136.5 -e LIST ALERTHISTORY
```

Command examples:

- **ACTIVEREQUEST**: Shows active requests currently being served by the storage servers.
- **ALERTDEFINITION**: Lists available storage-server alerts and their sources.
- **CELL**: Queries specific or all storage cell attributes (e.g., LIST CELL ATTRIBUTES A,B,C or ... ALL).
- **CELLDISK**: Lists storage cell disk attributes/details (e.g., LIST CELLDISK <name> DETAIL).
- **IORMPLAN**: Create/alter/drop/list IORM plans; view full details with LIST IORMPLAN DETAIL and apply plans to storage servers.
- **QUARANTINE**: Lists SQL statements prevented from using Smart Scan (e.g., LIST QUARANTINE DETAIL, supports WHERE).

Stay tuned



Operations Advisory
GitHub Repository



Thank you

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